

Transformer Architecture Review

We propose a new network architecture based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.

The scaled dot-product attention is defined as

$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / \sqrt{d_k}) V.$$

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.

Experiments show the model achieves 28.4 BLEU on English-to-German translation.